

Finite Gain Stabilization of Discrete-Time Linear Systems Subject to Actuator Saturation

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Abstract

It is shown that, for neutrally stable discrete-time linear systems subject to actuator saturation, finite gain l_p stabilization can be achieved by linear output feedback, for all $p \in (1, \infty]$. An explicit construction of the corresponding feedback laws is given. The feedback laws constructed also result in a closed-loop system that is globally asymptotically stable, and in an input-to-state estimate.

Key Words : input saturation, discrete-time linear systems, finite gain stability, Lyapunov functions.

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1 Introduction

In this paper, we consider the problem of global stabilization of a discrete-time linear system subject to actuator saturation:

$$\mathcal{P} : \begin{cases} x^+ = Ax + B\sigma(u + u_1), & x \in \mathbb{R}^n, u \in \mathbb{R}^m \\ y = Cx + u_2, & y \in \mathbb{R}^r \end{cases} \quad (1)$$

(we use the notation x^+ to indicate a forward shift, that is, for a function x and an integer t , $x^+(t)$ is $x(t+1)$), where $u_1 \in \mathbb{R}^m$ is the actuator disturbance, $u_2 \in \mathbb{R}^r$ is the sensor noise, and $\sigma : \mathbb{R}^m \rightarrow \mathbb{R}^m$ represents actuator saturation, i.e., $\sigma(s) = [\sigma_1(s_1) \ \sigma_2(s_2) \ \cdots \ \sigma_m(s_m)]$ with $\sigma_i(s_i) = \text{sign}(s_i) \min\{1, |s_i|\}$, and the pair (A, B) is stabilizable. The problem of global asymptotic stabilization (internal stabilization) of this system has recently been solved using nonlinear state feedback laws and under the condition that all the eigenvalues of A are inside or on the unit circle (Yang *et al.*, 1997), and for neutrally stable open-loop system using linear state feedback (Choi, 1999). Here, we are interested not only in closed-loop state space stability (internal stability), but also in stability with respect to both measurement and actuator noises. More specifically, we would like to construct a controller $\mathcal{C}$ so that the operator $(u_1, u_2) \mapsto (y_1, y_2)$ as defined by the following standard systems interconnection (see Fig. 1.)

$$\begin{cases} y_1 = \mathcal{P}(u_1 + y_2) \\ y_2 = \mathcal{C}(u_2 + y_1) \end{cases} \quad (2)$$

is well-defined and finite gain stable.

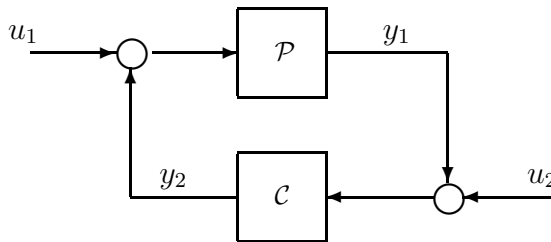


Figure 1: Standard closed-loop connection

We note that the disturbance u_1 we consider here is input additive and enters the system together with the control input u through the actuators. Simple examples show that the problem we consider does not always have a solution if the disturbance enters the system from outside the actuators.

The above problem was first studied for continuous-time systems. It was shown in (Liu *et al.*, 1996) that, for neutrally stable open loop systems, linear feedback laws can be used to achieve finite gain stability, with respect to every L_p -norm. For a neutrally stable system, all open loop poles are located in the closed left-half plane, with those on the $j\omega$ axis having Jordan blocks of size one. In the case that full state is available for feedback (i.e., $y_1 = x$ and $u_2 = 0$), it was shown in (Lin *et al.*, 1995) that if the external input signal is uniformly bounded, then finite-gain L_p -stabilization and local asymptotic stabilization can always be achieved simultaneously by linear feedback, no matter where the poles of the open loop system are. The uniform boundedness condition of (Lin *et al.*, 1995) was later removed by resorting to nonlinear feedback (Lin, 1997). Some other works related to the topic are Hou *et al.* (1997), Chitour *et al.* (1995), Nguyen and Jabbari (1997), Saberi *et al.* (1998), Suarez *et al.* (1997) and the references therein.

There are also several studies in the discrete-time setting, showing some of the continuous-time results carry over to discrete-time (for example, Hou *et al.*, 1997 and Yang *et al.*, 1997) and some do not (for

example, Hou *et al.*, 1997). In particular, Hou *et al.* (1997) shows that the results of Lin (1997) and Lin *et al.* (1995) on finite gain stabilization of continuous-time systems do not carry over to discrete-time systems. The objective of this paper is to show that the results of Liu *et al.* (1996), however, do carry over to discrete-time systems. More specifically, we show that, for neutrally stable discrete-time linear systems subject to actuator saturation, finite gain l_p stabilization can be achieved by linear output feedback for all $p \in (1, \infty]$. An explicit construction of the corresponding feedback laws is given. The feedback laws constructed also result in a closed-loop system that is globally asymptotically stable, and provide an input-to-state estimate. While many of the arguments used are conceptually similar to those used in the continuous-time case (Liu *et al.*, 1996), there are technical aspects that are very different and not totally obvious. For example, unlike in (Liu *et al.*, 1996), the feedback gain for the discrete-time case needs to be multiplied by a small factor, say κ , which causes the solution of a certain Lyapunov equation, and the subsequent estimation of the solution, to be dependent on κ (see Lemma 2). As another example, the difficulties in evaluating the difference of the non-quadratic Lyapunov function along the trajectories of the closed-loop system entail a careful estimation by Taylor expansion.

The remainder of the paper is organized as follows. Section 2 states the main results. Section 3 contains the proof of the results that were stated in Section 2. A brief concluding remark is given in Section 4.

2 Preliminary and Problem Statement

We first recall some notation. For a vector $X \in \mathbb{R}^\ell$, $|X|$ denotes the Euclidean norm of X , and for a matrix $X \in \mathbb{R}^{m \times n}$, the induced operator norm. For any $p \in [1, \infty)$, we write l_p^n for the set of all sequences $\{x(t)\}_{t=0}^\infty$, where $x \in \mathbb{R}^n$, such that $\sum_{t=0}^\infty |x(t)|^p < \infty$, and the l_p -norm of $x \in l_p^n$ is defined as $\|x\|_{l_p} = (\sum_{t=0}^\infty |x(t)|^p)^{1/p}$. We use l_∞^n to denote the set of all sequences $\{x(t)\}_{t=0}^\infty$, where $x \in \mathbb{R}^n$, such that $\sup_t |x(t)| < \infty$, and the l_∞ -norm of $x \in l_\infty^n$ is defined as $\|x\|_{l_\infty} = \sup_t |x(t)|$.

The objective of this paper is to show the following result concerning the global asymptotic stabilization as well as l_p -stabilization of system $\mathcal{P}$, as given by (1), using linear output feedback.

Theorem 1 *Consider a system (1). Let A be neutrally stable, i.e., all the eigenvalues of A are inside or on the unit circle, with those on the unit circle having all Jordan blocks of size one. Also assume that (A, B) is stabilizable and (A, C) is detectable. Then, there exists a linear observer-based output feedback law of the form*

$$\begin{cases} \hat{x}^+ = A\hat{x} + B\sigma(F\hat{x}) - L(y - C\hat{x}) \\ u = F\hat{x} \end{cases} \quad (3)$$

which has the following properties:

1. It is finite gain l_p -stable for all $p \in (1, \infty]$, i.e., there exists a $\gamma_p > 0$ such that

$$\|x\|_{l_p} \leq \gamma_p [\|u_1\|_{l_p} + \|u_2\|_{l_p}], \quad \forall u_1 \in l_p^m, u_2 \in l_p^r, \text{ and } x(0) = 0, \hat{x}(0) = 0. \quad (4)$$

2. In the absence of actuator and sensor noises u_1 and u_2 , the equilibrium $(x, \hat{x}) = (0, 0)$ is globally asymptotically stable.

Remark 1 *We will in fact actually obtain the following stronger ISS-like property (see, Sontag (1998) and references there):*

$$\|(x, \hat{x})\|_{l_p} \leq \theta_p (|x(0)| + |\hat{x}(0)|) + \gamma_p [\|u_1\|_{l_p} + \|u_2\|_{l_p}] \quad (5)$$

where θ_p is a class- $\mathcal{K}$ function. Observe that the single estimate (5) encompasses both the gain estimate (4) and asymptotic stability. Obviously, (4) is the special case of (5) for zero initial states. On the other hand, when applied with arbitrary initial states but $u_1 = u_2 = 0$, there follows that $(x, \hat{x})$ is in l_p , which implies, in particular, that $(x(t), \hat{x}(t))$ must converge to zero as $t \rightarrow \infty$ (global attraction) and that $|(x(t), \hat{x}(t))|$ is bounded by $\theta_p(|x(0)| + |\hat{x}(0)|)$ (stability).

3 Proof of Theorem 1

The proof of Theorem 1 will follow readily from the following proposition, which we establish first.

Proposition 1 *Let A be orthogonal (i.e., $A'A = I$), and suppose that the pair (A, B) is controllable. Then, the system*

$$x^+ = Ax + B\sigma(-\kappa B'Ax + u), \quad x \in \mathbb{R}^n, u \in \mathbb{R}^m \quad (6)$$

is finite gain l_p -stable, $p \in (1, \infty]$, for sufficiently small $\kappa > 0$. Moreover, for each $p \in (1, \infty]$ there exist a real γ_p , a $\kappa^ \in (0, 1]$, and a class- $\mathcal{K}$ function θ_p such that, for all $\kappa \in (0, \kappa^*]$,*

$$\|x\|_{l_p} \leq \gamma_p \|u\|_{l_p} + \theta_p(|x(0)|) \quad (7)$$

for all inputs $u \in l_p^m$ and all initial states $x(0)$.

To prove this proposition, we need to establish a few lemmas.

Lemma 1 *For any $p > l > 0$, there exist two scalars $M_1, M_2 > 0$ such that, for any two positive scalars ξ and ζ ,*

$$\xi^{p-l}\zeta^l \leq M_1\xi^p + M_2\zeta^p \quad (8)$$

and consequently, for any $n > 0$ and $\kappa > 0$,

$$\xi^{p-l}\zeta^l \leq M_1\kappa^n\xi^p + \kappa^{\frac{n(l-p)}{l}}M_2\zeta^p. \quad (9)$$

Proof of Lemma 1. Let $h : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ be defined as $h(x) = x^{\frac{l}{p-l}}$, which is continuous and strictly increasing with $h(0) = 0$ and $h(\infty) = \infty$, and $k(x) = x^{\frac{p-l}{l}}$ be its pointwise inverse. Define

$$H(x) = \int_0^x h(v)dv = \frac{p-l}{p}x^{\frac{p}{p-l}} \quad (10)$$

and

$$K(x) = \int_0^x k(v)dv = \frac{l}{p}x^{\frac{p}{l}}. \quad (11)$$

Letting $a = \xi^{p-l}$ and $b = \zeta^l$, it follows from Young's inequality (Hardy *et al.*, 1952), $ab \leq H(a) + K(b)$ for all $a, b \in \mathbb{R}^+$, that

$$\xi^{p-l}\zeta^l \leq \frac{p-l}{p}\xi^p + \frac{l}{p}\zeta^p = M_1\xi^p + M_2\zeta^p, \quad (12)$$

which also trivially implies (9). $\square$

Lemma 2 Let A and B be as given in Proposition 1. Then, for any $\kappa > 0$ such that $\kappa B'B < 2I$, $\bar{A}(\kappa) = A - \kappa BB'A$ is asymptotically stable. Moreover, let $P(\kappa)$ be the unique positive definite solution to the Lyapunov equation,

$$\bar{A}(\kappa)' P \bar{A}(\kappa) - P = -I. \quad (13)$$

Then, there exists a $\kappa^* > 0$ such that

$$\frac{\chi_1}{\kappa} I \leq P(\kappa) \leq \frac{\chi_2}{\kappa} I, \quad \forall \kappa \in (0, \kappa^*] \quad (14)$$

for some positive constants χ_1 and χ_2 independent of κ .

Proof of Lemma 2: The asymptotic stability of $\bar{A}$ follows from a simple Lyapunov/LaSalle argument (Choi, 1999). Let $\kappa_1^* > 0$ be such that $\kappa B'B < 2I$ for all $\kappa \in (0, \kappa_1^*]$. We recall that the solution to the Lyapunov equation (13) is given by

$$P(\kappa) = \sum_{k=0}^{\infty} (\bar{A}^k(\kappa))' \bar{A}^k(\kappa) = \sum_{k=0}^{\infty} [(A - \kappa BB'A)']^k [(A - \kappa BB'A)]^k. \quad (15)$$

Using the fact that $AA' = I$, we have

$$\begin{aligned} (A - \kappa BB'A)'(A - \kappa BB'A) &= I - 2\kappa A'BB'A + \kappa^2 A'BB'BB'A \\ &= I - \kappa A'B(2I - \kappa B'B)B'A. \end{aligned} \quad (16)$$

Using now the fact that $\kappa B'B < 2I$ for $\kappa \in (0, \kappa_1^*]$, we know that there exists $\kappa_2^* \in (0, \kappa_1^*]$ such that

$$\frac{1}{2}I \leq (A - \kappa BB'A)'(A - \kappa BB'A) \leq I, \quad \forall \kappa \in (0, \kappa_2^*]. \quad (17)$$

Again using the fact that $A'A = I$, we verify in a straightforward way that

$$(A - \kappa BB'A)^n = A^n - \kappa C_{A,B} C'_{A,B} A^n + \kappa^2 M_1(\kappa) \quad (18)$$

where $M_1(k)$ is a polynomial matrix in κ of order $n-2$, n being the order of the system (6), and

$$C_{A,B} = [B \quad AB \quad \dots \quad A^{n-1}B]$$

is the controllability matrix of the pair (A, B) and is of full rank. It then follows that

$$\begin{aligned} &((A - \kappa BB'A)^n)'(A - \kappa BB'A)^n \\ &= ((A^n)' - \kappa(A^n)'C_{A,B}C'_{A,B} + \kappa^2 M_1'(\kappa))(A^n - \kappa C_{A,B}C'_{A,B}A^n + \kappa^2 M_1(\kappa)) \\ &= I - 2\kappa(A^n)'C_{A,B}C'_{A,B}A^n + \kappa^2 M_2(\kappa) \end{aligned} \quad (19)$$

where $M_2(\kappa)$ is a symmetric polynomial matrix in κ of order $2n-2$. Since $C_{A,B}$ is of full rank, and because A is nonsingular, there exists a $\kappa^* \in (0, \kappa_2^*]$ such that

$$0 \leq I - \kappa M_1^0 I \leq ((A - \kappa BB'A)^n)'(A - \kappa BB'A)^n \leq I - \kappa M_2^0 I < I, \quad \forall \kappa \in (0, \kappa^*] \quad (20)$$

for some constants $M_1^0, M_2^0 > 0$ independent of κ .

Using (17), (20) and the fact that $A'A = I$ in (15), we have that for all $\kappa \in (0, \kappa^*]$,

$$P(\kappa) \leq \sum_{i=0}^{n-1} [(A - \kappa BB'A)^i]'(A - \kappa BB'A)^i \sum_{k=0}^{\infty} (1 - \kappa M_2^0)^k I \leq n \frac{1}{\kappa M_2^0} I = \frac{\chi_2}{\kappa} I \quad (21)$$

and

$$P(\kappa) \geq \sum_{i=0}^{n-1} [(A - \kappa BB'A)^i]' (A - \kappa BB'A)^i \sum_{k=0}^{\infty} (1 - \kappa M_1^0)^k I \geq \left(\frac{1}{2}\right)^{n-1} n \frac{1}{\kappa M_1^0} I = \frac{\chi_1}{\kappa} I \quad (22)$$

where $\chi_1 = \frac{n}{2^{n-1} M_1^0}$ and $\chi_2 = \frac{n}{M_2^0}$. $\square$

Lemma 3 *Let $\bar{A}(\kappa)$ be as given in Proposition 1, $P(\kappa)$ as defined in Lemma 2, then for any $p \in (1, \infty)$, there exists a $\kappa^* > 0$ such that*

$$[x' \bar{A}'(\kappa) P(\kappa) \bar{A}(\kappa) x]^{p/2} - [x' P(\kappa) x]^{p/2} \leq -\kappa^{\frac{2-p}{2}} \zeta |x|^p, \quad \kappa \in (0, \kappa^*), \quad (23)$$

where $\zeta > 0$ is some constant independent of κ .

Proof of Lemma 3. Inequality (23) holds trivially for $x = 0$. Hence in what follows, we assume, without loss of generality, that $x \neq 0$.

For simplicity, we introduce from now the following notation:

$$\mu = x' \bar{A}'(\kappa) P(\kappa) \bar{A}(\kappa) x \quad (24)$$

(where x and κ will be clear from the context). By the definition of $P(\kappa)$, we have

$$\mu - x' P(\kappa) x = -x' x. \quad (25)$$

From Lemma 2, there exists a $\kappa_1^* > 0$ such that for all $\kappa \in (0, \kappa_1^*]$

$$\left| \frac{x' x}{x' P(\kappa) x} \right| \leq \frac{4}{5}, \quad \forall x \neq 0. \quad (26)$$

With (25) and (26), we can continue the proof using Taylor expansion with remainder,

$$\begin{aligned} & [x' \bar{A}'(\kappa) P(\kappa) \bar{A}(\kappa) x]^{p/2} - [x' P(\kappa) x]^{p/2} \\ &= [x' P(\kappa) x - x' x]^{p/2} - [x' P(\kappa) x]^{p/2} \\ &\leq [x' P(\kappa) x]^{p/2} \left[1 - \frac{p}{2} \frac{x' x}{x' P(\kappa) x} + \delta \left(\frac{x' x}{x' P(\kappa) x} \right)^2 \right] - [x' P(\kappa) x]^{p/2} \\ &= -\frac{p}{2} [x' P(\kappa) x]^{\frac{p-2}{2}} |x|^2 + \delta [x' P(\kappa) x]^{\frac{p-4}{2}} |x|^4, \quad \kappa \in (0, \kappa_1^*) \end{aligned} \quad (27)$$

where $\delta = \max_{|z| \leq \frac{4}{5}} \left\{ \frac{p}{8} \left| (p-2)(1+z)^{\frac{p}{2}-2} \right| \right\}$ is a constant independent of κ .

Again by Lemma 2, there exists a $\kappa^* \in (0, \kappa_1^*)$ such that

$$[\mu]^{p/2} - [x' P(\kappa) x]^{p/2} \leq -\kappa^{\frac{2-p}{2}} \zeta |x|^p, \quad \kappa \in (0, \kappa^*) \quad (28)$$

for some $\zeta > 0$ independent of κ . $\square$

Lemma 4 *Let A and B be as given in Proposition 1. For any $l \in [1, \infty)$ and any $\kappa \in (0, 1]$,*

$$|\sigma(-\kappa B'A x + u)|^l \leq 2^{l-1} \kappa^l |B|^l |x|^l + 2^{l-1} |u|^l. \quad (29)$$

Proof of Lemma 4. Since σ is a standard saturation function and $|A| = 1$, for any $l \geq 1$, we have

$$|\sigma(-\kappa B'Ax + u)|^l \leq (\kappa|B||x| + |u|)^l \leq 2^{l-1}\kappa^l|B|^l|x|^l + 2^{l-1}|u|^l, \quad (30)$$

where the last inequality follows from Jensen's inequality applied to the convex function s^l :

$$(a + b)^l \leq \frac{1}{2}(2a)^l + \frac{1}{2}(2b)^l, \quad \forall a, b \geq 0.$$

□

Lemma 5 *Let A and B be as given in Proposition 1. Pick any $x \in \mathbb{R}^n$ and $u \in \mathbb{R}^m$, any number $\eta \geq 3$, and any nonnegative real number l . Denote $\tilde{x} = -\kappa B'Ax + u$. Then, provided $|x| > \eta|B\sigma(\tilde{x})|$, we have:*

$$|Ax + B\sigma(\tilde{x})|^l \leq |x|^l + l|x|^{l-2}x'A'B\sigma(\tilde{x}) + M|x|^{l-2}|B\sigma(\tilde{x})|^2, \quad (31)$$

for some constant $M > 0$ which is independent of κ .

Proof of Lemma 5. We first note that, since $|x| > \eta|B\sigma(\tilde{x})| \geq 3|B\sigma(\tilde{x})|$,

$$\frac{|2x'A'B\sigma(\tilde{x}) + |B\sigma(\tilde{x})|^2|}{|x|^2} \leq \frac{4}{5}. \quad (32)$$

Hence, using Taylor expansion with remainder, we have

$$\begin{aligned} |Ax + B\sigma(\tilde{x})|^l &= [|x|^2 + 2x'A'B\sigma(\tilde{x}) + |B\sigma(\tilde{x})|^2]^{\frac{l}{2}} \\ &= |x|^l \left(1 + \frac{2x'A'B\sigma(\tilde{x}) + |B\sigma(\tilde{x})|^2}{|x|^2} \right)^{\frac{l}{2}} \\ &\leq |x|^l \left[1 + \frac{l}{2} \frac{2x'A'B\sigma(\tilde{x}) + |B\sigma(\tilde{x})|^2}{|x|^2} + \delta \left(\frac{2x'A'B\sigma(\tilde{x}) + |B\sigma(\tilde{x})|^2}{|x|^2} \right)^2 \right] \\ &\leq |x|^l + l|x|^{l-2}x'A'B\sigma(\tilde{x}) + \frac{l}{2}|x|^{l-2}|B\sigma(\tilde{x})|^2 + \delta|x|^{l-4}((2 + 1/\eta)|x||B\sigma(\tilde{x})|)^2 \end{aligned} \quad (33)$$

where $\delta = \max_{|z| \leq \frac{4}{5}} \left\{ \frac{l}{8} \left| (l-2)(1+z)^{\frac{l}{2}-2} \right| \right\}$ is a constant independent of κ .

So we can see that the inequality (31) holds for $M = \frac{l}{2} + \delta(2 + 1/\eta)^2$. □

We are now ready to prove Proposition 1.

Proof of Proposition 1: We separate the proof for $p \in (1, \infty)$ and for $p = \infty$.

Proof for $p \in (1, \infty)$.

For clarity, let us repeat here the system equation (6):

$$x^+ = Ax + B\sigma(-\kappa B'Ax + u), \quad x \in \mathbb{R}^n, u \in \mathbb{R}^m. \quad (34)$$

This may also be rewritten as:

$$x^+ = \bar{A}(\kappa)x + B(-\tilde{x} + \sigma(\tilde{x}) + u), \quad x \in \mathbb{R}^n, u \in \mathbb{R}^m, \quad (35)$$

where $\bar{A}(\kappa) = A - \kappa BB'A$, $\tilde{x} = -\kappa B'Ax + u$.

For this system, define the function V_1 as:

$$V_1(x) = (x'P(\kappa)x)^{p/2}, \quad (36)$$

where $P(\kappa)$ is as given in Lemma 2. We next evaluate the increments $V(x^+(t)) - V(x(t))$, which we denote as “ ΔV_1 ” for short, along any given trajectory of (35). It is convenient to treat separately the cases $|x| > \eta|B\sigma(\tilde{x})|$ and $|x| \leq \eta|B\sigma(\tilde{x})|$. Here $\eta \geq 3$ is a number to be specified soon.

Case 1: $|x| > \eta|B\sigma(\tilde{x})|$.

Using the definition of V_1 , we now give an upper bound on ΔV_1 along the trajectories of the system (35). To simplify the equations, we introduce the following notation:

$$\nu = 2\kappa x' A' P(\kappa) B B' A x + 2x' A' P(\kappa) B \sigma(\tilde{x}) + \sigma'(\tilde{x}) B' P(\kappa) B \sigma(\tilde{x}),$$

in addition to μ as defined in Equation (24). Thus:

$$\begin{aligned} \Delta V_1 &= V_1^+ - V_1 \\ &= [(x^+)' P(\kappa) x^+]^{p/2} - [x' P(\kappa) x]^{p/2} \\ &= [\bar{A}(\kappa)x + \kappa B B' A x + B \sigma(\tilde{x})]' P(\kappa) [\bar{A}(\kappa)x + \kappa B B' A x + B \sigma(\tilde{x})]^{p/2} - [x' P(\kappa) x]^{p/2} \\ &= [\mu + 2x' A' P(\kappa) B (\kappa B' A x + \sigma(\tilde{x})) - \kappa^2 x' A' B B' P(\kappa) B B' A x \\ &\quad + \sigma'(\tilde{x}) B' P(\kappa) B \sigma(\tilde{x})]^{p/2} - [x' P(\kappa) x]^{p/2} \\ &\leq [\mu + \nu]^{p/2} - [x' P(\kappa) x]^{p/2} \\ &= [\mu]^{p/2} \left[1 + \frac{\nu}{\mu} \right]^{p/2} - [x' P(\kappa) x]^{p/2}. \end{aligned} \tag{37}$$

By Lemma 2, there exist a $\kappa_1^* > 0$ and $\eta \geq 3$ independent of κ , such that for all $|x| > \eta|B\sigma(\tilde{x})|$

$$\left| \frac{\nu}{\mu} \right| \leq \frac{4}{5}, \quad \kappa \in (0, \kappa_1^*). \tag{38}$$

To see this, let $\kappa_0^* > 0$ be such that (14) of Lemma 2 and (17) in the proof of Lemma 2 both hold for all $\kappa \in (0, \kappa_0^*]$. Then, for all $\kappa \in (0, \kappa_0^*]$, we have

$$|\nu| \leq \left[2\chi_2 |B|^2 + \frac{2\chi_2}{\kappa\eta} + \frac{\chi_2}{\kappa\eta^2} \right] |x|^2 \tag{39}$$

and

$$|\mu| \geq \frac{\chi_1}{2\kappa} |x|^2, \tag{40}$$

from which it is clear that there exist κ_1^* and $\eta > 3$ such that (38) holds.

Next, we may use a Taylor expansion with remainder to continue the bounding of ΔV_1 as follows:

$$\Delta V_1 \leq [\mu]^{p/2} \left[1 + \frac{p}{2} \frac{\nu}{\mu} + \delta \left[\frac{\nu}{\mu} \right]^2 \right] - [x' P(\kappa) x]^{p/2} \tag{41}$$

where $\delta = \max_{|z| \leq \frac{4}{5}} \left\{ \frac{p}{8} \left| (p-2)(1+z)^{\frac{p}{2}-2} \right| \right\}$ is a constant independent of κ .

By Lemma 3 and Lemma 2, there exists $\kappa_2^* \in (0, \kappa_1^*]$ such that for any $\kappa \in (0, \kappa_2^*]$, we have

$$\begin{aligned} \Delta V_1 &\leq -\kappa^{\frac{2-p}{2}} \zeta |x|^p + \frac{p}{2} [\mu]^{\frac{p-2}{2}} [\nu] + \delta [\mu]^{\frac{p-4}{2}} [\nu]^2 \\ &\leq -\kappa^{\frac{2-p}{2}} \zeta |x|^p + \psi_1 \kappa^{\frac{2-p}{2}} |x|^{p-2} [2|x||P(\kappa)B||\tilde{x} - \sigma(\tilde{x})| + 2|x||P(\kappa)B||u| + |P(\kappa)||B\sigma(\tilde{x})|^2] \\ &\quad + \psi_2 \kappa^{\frac{4-p}{2}} |x|^{p-4} [2|x|^2 |\kappa P(\kappa)||B|^2 + 2|P(\kappa)||x||B\sigma(\tilde{x})| + |P(\kappa)||B\sigma(\tilde{x})|^2]^2 \end{aligned} \tag{42}$$

where $\zeta > 0$ is as defined in Lemma 3, and $\psi_1, \psi_2 > 0$ are some constants independent of κ .

Before continuing, we digress to observe that

$$|\tilde{x} - \sigma(\tilde{x})| \leq \tilde{x}'\sigma(\tilde{x}). \quad (43)$$

Using (43), Lemma 1, Lemma 4, and the condition $|x| > \eta|B\sigma(\tilde{x})|$, we can show that there exists a $\kappa_3^* \in (0, \kappa_2^*]$ such that for all $\kappa \in (0, \kappa_3^*]$ the estimation of ΔV_1 can be now concluded as follows:

$$\Delta V_1 \leq -\kappa^{\frac{2-p}{2}} \zeta |x|^p + 2\psi_1 \kappa^{\frac{2-p}{2}} |x|^{p-1} |P(\kappa)B|\tilde{x}'\sigma(\tilde{x}) + M_{1a}\kappa^{\frac{2-p}{2}} \max\{\kappa, \kappa^{p-1}\}|x|^p + M_{2a}(\kappa)|u|^p, \quad (44)$$

where $M_{1a} > 0, M_{2a}(\kappa) > 0$ with M_{1a} independent of κ are defined in an obvious way. In deriving (44), we have also used the fact that $|x|^{p-2} < (|B\sigma(\tilde{x})|/\eta)^{p-2}$ for $p < 2$ and $B\sigma(\tilde{x}) \neq 0$.

Case 2: $|x| \leq \eta|B\sigma(\tilde{x})|$.

By using Lemma 2, Lemma 3 and Lemma 4, ΔV_1 along the trajectories of (35) is bounded as follows:

$$\begin{aligned} \Delta V_1 &= [(x^+)'P(\kappa)x^+]^{p/2} - [x'P(\kappa)x]^{p/2} \\ &\leq |P(\kappa)|^{p/2} |Ax + B\sigma(\tilde{x})|^p - [x'P(\kappa)x]^{p/2} + [\mu]^{p/2} \\ &\leq -\kappa^{\frac{2-p}{2}} \zeta |x|^p + \kappa^{-\frac{p}{2}} \chi_2^{\frac{p}{2}} (|x| + |B\sigma(\tilde{x})|)^p \\ &\leq -\kappa^{\frac{2-p}{2}} \zeta |x|^p + (\eta + 1)^p \kappa^{-\frac{p}{2}} \chi_2^{\frac{p}{2}} |B\sigma(\tilde{x})|^p \\ &\leq -\kappa^{\frac{2-p}{2}} \zeta |x|^p + M_{1b} \kappa^{-\frac{p-2}{2}} \kappa^{p-1} |x|^p + M_{2b}(\kappa) |u|^p, \quad \kappa \in (0, \kappa_3^*] \end{aligned} \quad (45)$$

where $\chi_2 > 0$ and $\zeta > 0$ are as defined in Lemma 2 and Lemma 3 respectively, and $M_{1b} > 0, M_{2b}(\kappa) > 0$ are constants with M_{1b} being independent of κ .

Summarizing, we may combine Case 1 with Case 2, to obtain:

$$\Delta V_1 \leq \begin{cases} -\kappa^{\frac{2-p}{2}} \zeta |x|^p + 2\psi_1 \kappa^{\frac{2-p}{2}} |P(\kappa)B||x|^{p-1} \tilde{x}'\sigma(\tilde{x}) \\ \quad + M_1 \kappa^{\frac{2-p}{2}} \max\{\kappa, \kappa^{p-1}\} |x|^p + M_2(\kappa) |u|^p, & \text{if } |x| > \eta|B\sigma(\tilde{x})|, \\ -\kappa^{\frac{2-p}{2}} \zeta |x|^p + M_1 \kappa^{\frac{2-p}{2}} \kappa^{p-1} |x|^p + M_2(\kappa) |u|^p, & \text{if } |x| \leq \eta|B\sigma(\tilde{x})|, \end{cases} \quad (46)$$

where $M_1 = \max\{M_{1a}, M_{1b}\}$ and $M_2(\kappa) = \max\{M_{2a}(\kappa), M_{2b}(\kappa)\}$.

For system (34), we next define another function:

$$V_0(x) = |x|^{p+1}. \quad (47)$$

An estimation of its increments along the trajectories of (34) can also be carried out by separately considering each of the cases $|x| > \eta|B\sigma(\tilde{x})|$ and $|x| \leq \eta|B\sigma(\tilde{x})|$.

Case 1: $|x| \leq \eta|B\sigma(\tilde{x})|$.

By Lemma 4, for any $\kappa \in (0, \kappa_3^*]$,

$$\begin{aligned} \Delta V_0 &= |Ax + B\sigma(\tilde{x})|^{p+1} - |x|^{p+1} \leq |Ax + B\sigma(\tilde{x})|^{p+1} \\ &\leq (|x| + |B\sigma(\tilde{x})|)^{p+1} \leq ((\eta + 1)|B\sigma(\tilde{x})|)^{p+1} \leq \kappa N_{1a} |x|^p + N_{2a} |u|^p \end{aligned} \quad (48)$$

for some positive constants N_{1a} and N_{2a} independent of κ . In deriving (48), we have used the fact that both σ and κ are bounded.

Case 2: $|x| > \eta|B\sigma(\tilde{x})|$.

By Lemma 5, Lemma 4 and Lemma 1, there exists $\kappa_4^* \in (0, \kappa_3^*]$ such that for any $\kappa \in (0, \kappa_4^*]$,

$$\begin{aligned}\Delta V_0 &= |Ax + B\sigma(\tilde{x})|^{p+1} - |x|^{p+1} \\ &\leq |x|^{p+1} + (p+1)|x|^{p-1}x'A'B\sigma(\tilde{x}) + N_{1b}|B\sigma(\tilde{x})|^2|x|^{p-1} - |x|^{p+1} \\ &\leq -\frac{p+1}{\kappa}|x|^{p-1}\tilde{x}^T\sigma(\tilde{x}) + \kappa N_{1c}|x|^p + N_{2c}(\kappa)|u|^p\end{aligned}\quad (49)$$

where $N_{1c}, N_{1b} > 0, N_{2c}(\kappa) > 0$ are constants, and N_{1b}, N_{1c} are independent of κ . In deriving (49), the first inequality by Lemma 5, the second inequality is the consequence of the fact that σ is bounded and Lemmas 4 and 1.

Combining Case 1 with Case 2, we have, for any $\kappa \in (0, \kappa_4^*]$:

$$\Delta V_0 \leq \begin{cases} -\frac{p+1}{\kappa}|x|^{p-1}\tilde{x}^T\sigma(\tilde{x}) + \kappa N_1|x|^p + N_2(\kappa)|u|^p, & \text{if } |x| > \eta|B\sigma(\tilde{x})|, \\ \kappa N_1|x|^p + N_2(\kappa)|u|^p, & \text{if } |x| \leq \eta|B\sigma(\tilde{x})|, \end{cases}\quad (50)$$

where $N_1 = \max\{N_{1a}, N_{1c}\}$ and $N_2(\kappa) = \max\{N_{2a}, N_{2c}(\kappa)\}$.

Finally, we define the following Lyapunov (or “storage”) function:

$$V(x) = V_1(x) + \varpi V_0(x), \quad (51)$$

where

$$\varpi = \frac{2}{p+1}\kappa^{\frac{4-p}{2}}\psi_1|P(\kappa)B|.$$

It is straightforward to verify that there exists some $\kappa^* \in (0, \kappa_4^*]$ such that

$$\Delta V(x) \leq -\kappa^{\frac{2-p}{2}}\alpha|x|^p + \beta(\kappa)|u|^p, \quad \forall \kappa \in (0, \kappa^*] \quad (52)$$

for some $\alpha \in (0, \zeta)$ and $\beta(\kappa) > 0$.

Now consider an arbitrary initial state $x(0)$ and control u , and the ensuing trajectory x . Summing both sides of (52) from $t = 0$ to ∞ and using the fact that V is nonnegative, we conclude that:

$$\kappa^{\frac{2-p}{2}}\alpha\|x\|_{l_p}^p \leq \beta(\kappa)\|u\|_{l_p}^p + \theta_{p0}(|x(0)|), \quad (53)$$

where $\theta_{p0}(r) = \varpi r^{p+1} + \left(\frac{\chi_2 r^2}{\kappa}\right)^{p/2}$. This implies that

$$\|x\|_{l_p} \leq \gamma_p\|u\|_{l_p} + \theta_p(|x(0)|), \quad (54)$$

where $\gamma_p = \left(\kappa^{\frac{p-2}{2}}\beta(\kappa)/\alpha\right)^{\frac{1}{p}}$, and $\theta_p(r) = \left(\kappa^{\frac{p-2}{2}}\theta_{p0}(r)/\alpha\right)^{\frac{1}{p}}$.

Proof for $p = \infty$.

From (52) we get for $p = 2$,

$$\Delta V(x) \leq -\alpha|x|^2 + \beta(\kappa)\|u\|_{l_\infty}^2. \quad (55)$$

Hence, $\Delta V(x)$ is negative outside the ball of radius $(\beta(\kappa)/\alpha)^{\frac{1}{2}}\|u\|_{l_\infty}$ centered at the origin, from which it follows that, for any state $x(t)$ in the trajectory:

$$V(x(t)) \leq \left(\frac{\varpi\beta^{\frac{3}{2}}(\kappa)}{\alpha^{\frac{3}{2}}}\|u\|_{l_\infty} + \frac{\chi_2\beta(\kappa)}{\alpha\kappa}\right)\|u\|_{l_\infty}^2 + \theta_{\infty 0}(|x(0)|), \quad (56)$$

where $\theta_{\infty 0}(r) = \frac{\chi_2}{\kappa}r^2 + \varpi r^3$. If $\|u\|_{l_\infty} \leq 1$, we have:

$$\frac{\chi_1}{\kappa}|x(t)|^2 \leq x(t)'P(\kappa)x(t) \leq V(x(t)) \quad (57)$$

and

$$V(x(t)) \leq \left(\frac{\varpi \beta^{\frac{3}{2}}(\kappa)}{\alpha^{\frac{3}{2}}} + \frac{\chi_2 \beta(\kappa)}{\alpha \kappa} \right) \|u\|_{l_\infty}^2 + \theta_{\infty 0}(|x(0)|), \quad (58)$$

which implies the following estimate for the entire trajectory:

$$\|x\|_{l_\infty} \leq \left\{ \frac{\kappa \varpi \beta^{\frac{3}{2}}(\kappa)}{\alpha^{\frac{3}{2}} \chi_1} + \frac{\chi_2 \beta(\kappa)}{\alpha \chi_1} \right\}^{\frac{1}{2}} \|u\|_{l_\infty} + \theta_{\infty 1}(|x(0)|), \quad (59)$$

where $\theta_{\infty 1}(r) = \left(\frac{\kappa \theta_{\infty 0}(r)}{\chi_1} \right)^{\frac{1}{2}}$. If, instead, $\|u\|_{l_\infty} > 1$, we have:

$$\varpi |x|^3 \leq V(x) \leq \left(\frac{\varpi \beta^{\frac{3}{2}}(\kappa)}{\alpha^{\frac{3}{2}}} + \frac{\chi_2 \beta(\kappa)}{\alpha \kappa} \right) \|u\|_{l_\infty}^3 + \theta_{\infty 1}(|x(0)|), \quad (60)$$

from which we get that

$$\|x\|_{l_\infty} \leq \left(\frac{\beta^{\frac{3}{2}}(\kappa)}{\alpha^{\frac{3}{2}} \varpi} + \frac{\chi_2 \beta(\kappa)}{\kappa \alpha \varpi} \right)^{\frac{1}{3}} \|u\|_{l_\infty} + \theta_{\infty 2}(|x(0)|), \quad (61)$$

where $\theta_{\infty 2}(r) = \left(\frac{\theta_{\infty 0}(r)}{\varpi} \right)^{\frac{1}{3}}$. Letting

$$\gamma_\infty = \max \left\{ \left\{ \frac{\kappa \varpi \beta^{\frac{3}{2}}(\kappa)}{\alpha^{\frac{3}{2}} \chi_1} + \frac{\chi_2 \beta(\kappa)}{\alpha \chi_1} \right\}^{\frac{1}{2}}, \left\{ \frac{\beta^{\frac{3}{2}}(\kappa)}{\alpha^{\frac{3}{2}} \varpi} + \frac{\chi_2 \beta(\kappa)}{\kappa \alpha \varpi} \right\}^{\frac{1}{3}} \right\}$$

and $\theta_\infty = \max\{\theta_{\infty 1}, \theta_{\infty 2}\}$, we have, finally, the required conclusion:

$$\|x\|_{l_\infty} \leq \gamma_\infty \|u\|_{l_\infty} + \theta_\infty(|x(0)|) \quad (62)$$

for $p = \infty$ as well. $\square$

We are now ready to prove Theorem 1.

Proof of Theorem 1. Without loss of generality, making a change of coordinates if required, we may assume that the system (1) has the following partitioned form:

$$\begin{cases} \dot{x}_1^+ = A_1 x_1 + B_1 \sigma(u + u_1) \\ \dot{x}_0^+ = A_0 x_0 + B_0 \sigma(u + u_1) \\ y = Cx + u_2 \end{cases} \quad (63)$$

where A_1 is orthogonal and A_0 is asymptotically stable, and

$$A = \begin{bmatrix} A_1 & 0 \\ 0 & A_0 \end{bmatrix}, \quad B = \begin{bmatrix} B_1 \\ B_0 \end{bmatrix}.$$

We construct the output feedback law in the form of (3) with $F = [-\kappa B_1' A_1 \quad 0]$, the matrix L being chosen such that $A + LC$ is asymptotically stable. Using this feedback, the closed-loop system is:

$$\begin{cases} \dot{x}_1^+ = A_1 x_1 + B_1 \sigma(-\kappa B_1' A_1 \hat{x}_1 + u_1) \\ \dot{x}_0^+ = A_0 x_0 + B_0 \sigma(-\kappa B_1' A_1 \hat{x}_1 + u_1) \\ \hat{x}^+ = A\hat{x} + B\sigma(-\kappa B_1' A_1 \hat{x}_1) - L(Cx - C\hat{x} + u_2). \end{cases} \quad (64)$$

Let $e = [e'_1 \quad e'_0]'$, where $e_1 = x_1 - \hat{x}_1$ and $e_0 = x_0 - \hat{x}_0$. Here we have partitioned $\hat{x} = [\hat{x}'_1 \quad \hat{x}'_0]'$ accordingly. In the new states (x, e) , (64) can be written as follows,

$$\begin{cases} x_1^+ = A_1 x_1 + B_1 \sigma(-\kappa B'_1 A_1 x_1 + \kappa B'_1 A_1 e_1 + u_1) \\ x_0^+ = A_0 x_0 + B_0 \sigma(-\kappa B'_1 A_1 x_1 + \kappa B'_1 A_1 e_1 + u_1) \\ e^+ = (A + LC)e \\ \quad + B[\sigma(-\kappa B'_1 A_1 x_1 + \kappa B'_1 A_1 e_1 + u_1) - \sigma(-\kappa B'_1 A_1 x_1 + \kappa B'_1 A_1 e_1)] + Lu_2. \end{cases} \quad (65)$$

Since σ is global Lipschitz with a Lipschitz constant 1,

$$|\sigma(-\kappa B'_1 A_1 x_1 + \kappa B'_1 A_1 e_1 + u_1) - \sigma(-\kappa B'_1 A_1 x_1 + \kappa B'_1 A_1 e_1)| \leq |u_1|. \quad (66)$$

Noting that $A + LC$ is asymptotically stable and viewing

$$\sigma(-\kappa B'_1 A_1 x_1 + \kappa B'_1 A_1 e_1 + u_1) - \sigma(-\kappa B'_1 A_1 x_1 + \kappa B'_1 A_1 e_1) + Lu_2$$

as an l_p input to the e -subsystem, we have that, for some constant $\gamma_{pe} > 0$,

$$\|e\|_{l_p} \leq \gamma_{pe}(\|u_1\|_{l_p} + \|u_2\|_{l_p} + |e(0)|). \quad (67)$$

Next, applying Proposition 1 to the x_1 -subsystem, and viewing $\kappa B'_1 A_1 e_1 + u_1$ as an l_p input to this subsystem, we have,

$$\|x_1\|_{l_p} \leq \gamma_{p1}(\|u_1\|_{l_p} + \|u_2\|_{l_p} + |e(0)|) + \theta_{p1}(|x_1(0)|)$$

for some $\gamma_{p1} > 0$ and θ_{p1} of class $\mathcal{K}$.

On the other hand, viewing $\sigma(-\kappa B'_1 A_1 x_1 + \kappa B'_1 A_1 e_1 + u_1)$ as an l_p input to the x_0 -subsystem, we have the estimate:

$$\|x_0\|_{l_p} \leq \gamma_{p0}(\|x_1\|_{l_p} + \|e\|_{l_p} + \|u_1\|_{l_p} + |x_0(0)|),$$

for some $\gamma_{p0} > 0$.

In conclusion, we have,

$$\begin{aligned} \|x\|_{l_p} &\leq \|x_1\|_{l_p} + \|x_0\|_{l_p} \\ &\leq \gamma_p(\|u_1\|_{l_p} + \|u_2\|_{l_p}) + \varphi_p(|e(0)| + |x(0)|) \end{aligned} \quad (68)$$

where $\gamma_p > 0$ is some constant and φ_p is a suitable class- $\mathcal{K}$ function. Together with (67), and changing back to the original coordinates, we also conclude that an estimate like the one in (5) holds. $\square$

4 Conclusions

In this paper, we have established that a discrete-time, neutrally stable, stabilizable, and detectable linear system, when subject to actuator saturation, is finite gain l_p stabilizable by linear feedback, for any $p \in (1, \infty]$. A linear output feedback law which simultaneously achieves l_p stabilization and global asymptotic stabilization was constructed.

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